



Adhoc & Sensor Networks unit 4

Adhoc & Sensor Networks (Jawaharlal Nehru Technological University, Hyderabad)



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UNIT-4

Basics of Wireless Sensors and Lower Layer Issues – Simplified Explanation

Wireless Sensor Networks (WSNs) are made up of tiny, low-power devices called **sensor nodes** that can collect and send data wirelessly. These networks combine sensing, computing, and communication to monitor environments in real-time.

Introduction

In recent years, improvements in small electronic components, low-power circuit designs, and affordable batteries have made it possible to develop **Wireless Sensor Networks (WSNs)**. These networks combine small wireless devices with sensing and communication capabilities to monitor physical environments.

Example:

Think of a **smart home** where temperature sensors adjust the air conditioning based on room temperature.

Each sensor in a WSN is small, about the size of a few cubic millimeters, and costs less than **one US dollar**. These sensors have a radio, microcontroller, battery, and a sensing unit. The main function of a sensor is to **detect changes** in its surroundings and send this information to a user.

WSNs are different from traditional networks because they focus on **data-centric communication**. Instead of sending information from one specific node, the system gathers data based on **attributes** like:

- “Which area has a temperature above 35°C?”
- “Where is the noise level above 70 dB?”

Since many sensors need to be deployed to ensure accurate readings, **WSNs work best with large numbers of small sensors**.

Functional Components of a Sensor

A typical **sensor node** consists of the following parts:

1. **Transducer** – Converts a physical quantity (like temperature or pressure) into an electrical signal.
2. **Embedded Processor** – Processes the collected data.
3. **Memory Unit** – Stores sensor data.
4. **Wireless Transceiver** – Sends and receives information.
5. **Battery** – Provides power to the sensor.

Example:

A **smart farm** uses soil moisture sensors to detect water levels and automatically activate irrigation when needed. The **transducer** detects moisture, the **processor** decides if irrigation is needed, and the **wireless transceiver** sends an alert to the farmer's phone.

Each sensor type depends on the application. Some **off-the-shelf sensors** are available for specific tasks, while **custom sensors** can be built by connecting transducers to external devices.

Design Issues

The latest advancements in technology allow hundreds or even thousands of tiny, **low-power wireless sensors** to be deployed for data collection.

Advantages of WSNs Over Wired Networks

1. **Easy Deployment** – Wireless sensors can be placed **without cables**, reducing installation costs.
 - *Example: Sensors dropped from a plane to track forest fires.*
2. **Extended Range** – Instead of using one large wired sensor, multiple wireless sensors can cover a **larger area**.
 - *Example: Multiple water sensors in a city can detect leaks faster than a single central unit.*
3. **Fault Tolerance** – If one sensor **fails**, others continue working.
 - *Example: If one temperature sensor in a greenhouse stops working, others still monitor the environment.*
4. **Mobility** – Wireless sensors can be **moved** if necessary.
 - *Example: Sensors attached to robots in disaster zones adjust their position to collect more accurate data.*

However, wireless networks have some **limitations**, such as **low bandwidth, signal interference, and limited battery life**. Since replacing batteries is difficult in remote areas, **energy-efficient protocols** must be used to extend the network's lifespan.

Routing Protocol Challenges

Traditional **Mobile Ad Hoc Network (MANET)** routing protocols do not work well for WSNs because:

- **Data-Centric Requests** – Information is gathered based on conditions, not from specific nodes.
 - *Example: Instead of asking for data from "Sensor A," the system requests "all temperatures above 35°C."*
- **No Unique Node ID** – Unlike traditional networks, WSNs do not require each node to have a unique identity (like an IP address).

- **Data Aggregation** – Similar data from nearby sensors is combined before transmission to save energy.
 - *Example: Instead of sending 10 identical temperature readings, a summary report is sent instead.*
- **Application-Specific** – Different applications require different routing techniques.
 - *Example: A surveillance sensor network works differently from a weather monitoring system.*

Key Features of an Ideal Sensor Network

1. **Attribute-Based Addressing** – Sensors respond to specific requests based on attributes like location and condition.
 - *Example: "Send data from all sensors in Hyderabad where the temperature is above 40°C."*
 2. **Location Awareness** – Sensors should know their position for accurate data collection.
 - *Example: GPS-based animal tracking collars use location awareness to report wildlife movements.*
 3. **Real-Time Response** – Some applications require immediate alerts when conditions change.
 - *Example: A security system must send an instant alert if an unauthorized person enters a restricted area.*
 4. **Efficient Query Handling** – Users should be able to request and receive data quickly through a base station or any nearby sensor.
 - *Example: A health monitoring system should immediately notify doctors if a patient's oxygen level drops below 90%.*
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Challenges

WSNs face several **technical limitations**, including:

1. **Limited Energy** – Since sensors run on batteries, **power-saving techniques** are necessary.
2. **Self-Organizing** – The network should automatically adjust if sensors fail or new ones are added.
3. **Stationary vs. Mobile Sensors** – Some sensors stay in one place, while others (e.g., on robots) may move slightly.
4. **Redundant Data** – Since nearby sensors detect similar information, data **aggregation** is necessary to avoid sending duplicate data.
5. **High Sensor Density** – WSNs consist of thousands of sensors, making individual node IDs impractical.

Example:

- In **smart agriculture**, thousands of sensors monitor soil conditions. If one sensor fails, nearby sensors ensure data accuracy.
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Lower Layer Issues in WSNs

1. **Ad Hoc Deployment** – Since sensors are randomly placed, the network must **self-adjust** and create connections automatically.
2. **Limited Computing Power** – Sensors cannot run complex algorithms, so **lightweight protocols** are used.
3. **Energy Conservation** – Sensors must **balance energy use** and accuracy to extend network life.
4. **Scalability** – WSNs must handle thousands of nodes efficiently.
5. **Short Communication Range** – Wireless sensors have **limited bandwidth** and require multi-hop communication.
6. **Fault Tolerance** – If sensors fail, the system should **adjust routes** and continue functioning.
7. **Connectivity Issues** – Sensors should remain connected despite **failures or environmental changes**.
8. **Security Concerns** – Data encryption is needed but must be **energy-efficient**.
9. **Quality of Service (QoS)** – Time-sensitive applications (e.g., **military or healthcare**) require fast data transmission.
10. **Overhead Costs** – Frequent retransmissions due to interference increase **energy consumption**.

Example:

- In a **disaster response system**, sensors detecting fire must immediately alert rescue teams. Any delay due to network congestion can be dangerous.
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Conclusion

WSNs are **powerful tools** for real-time monitoring but require **efficient energy use, reliable data transmission, and security mechanisms**. Unlike traditional networks, **data-centric communication, self-organizing capabilities, and scalability** are crucial for WSN performance. Future advancements will continue improving **battery life, connectivity, and sensor accuracy** to expand WSN applications across various fields.

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Applications – Simplified Explanation

Wireless Sensor Networks (WSNs) are widely used in various fields to monitor conditions in real-time and send alerts if a problem is detected.

Monitoring in Structures

Thousands of sensors can be placed inside and outside structures like **cars, airplanes, and buildings** to constantly monitor their condition. If a major issue is detected, a **real-time warning** is issued.

Example:

- In **airplanes**, sensors monitor engine temperature, pressure, and vibrations. If anything unusual is detected, pilots receive a warning.
- In **cars**, tire pressure sensors alert drivers if a tire is losing air.

However, traditional **wired sensors** used for monitoring are **large, expensive, and need a continuous power supply**. These sensors require a **pre-planned setup**, meaning strategic positions must be chosen before installation.

A major issue with **wired sensors** is that **if one sensor fails, the whole system might stop working**, leaving parts of the structure unmonitored. This is a **big problem** if sensors are placed in **hard-to-reach locations** like inside bridges or deep underground.

Example:

- If a **temperature sensor in an aircraft engine fails**, it may prevent early detection of overheating, leading to serious damage.
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Military Applications

WSNs were originally developed for **military use**, especially for:

1. **Battlefield Surveillance** – Sensors track enemy movement and send real-time alerts.
2. **Enemy Tracking** – Sensors detect footsteps, vehicle movement, and gunfire in war zones.

Example:

- In **border security**, wireless sensors can detect unauthorized movements and alert soldiers.
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Civil Applications

Although WSNs were first designed for military use, they are now **more commonly used in civilian applications**. These networks help monitor and control **various environments** like cities, industries, and homes.

Example:

- **Smart cities** use WSNs to monitor traffic, pollution, and weather conditions.

- **Agriculture** uses soil sensors to track moisture levels and automate irrigation.
- **Healthcare** uses wireless sensors to track **heart rate, blood pressure, and glucose levels** in patients.

Environmental Monitoring

Wireless sensors can be placed in forests, rivers, and cities to monitor environmental conditions and detect natural disasters.

Examples:

- **Forest Fire Detection** – Sensors measure temperature and smoke levels to detect wildfires early and send alerts to firefighters.
 - **Air Pollution Monitoring** – Sensors track harmful gases like CO₂ and NO₂ in cities and industrial areas to improve air quality.
 - **Flood Detection** – Water level sensors in rivers send alerts if flooding is likely, helping authorities take preventive measures.
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Healthcare and Medical Applications

Wireless sensors are used in **hospitals, homes, and wearable devices** to monitor patients' health in real time.

Examples:

- **Remote Patient Monitoring** – Wearable sensors track heart rate, blood pressure, and oxygen levels and send reports to doctors.
 - **Diabetes Management** – Wireless glucose sensors monitor blood sugar levels and alert patients if intervention is needed.
 - **Elderly Care** – Motion sensors detect falls in senior citizens and automatically notify caregivers.
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Agriculture and Precision Farming

WSNs help farmers improve crop production by providing **real-time data** about soil, weather, and plant health.

Examples:

- **Soil Moisture Monitoring** – Sensors track soil moisture levels and automatically turn on irrigation systems.
 - **Pest Detection** – Sensors detect harmful insects and trigger alerts for pesticide spraying.
 - **Livestock Tracking** – Wireless collars monitor cattle movements and health conditions, reducing losses.
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Smart Homes and Buildings

Wireless sensors enhance security, automation, and energy efficiency in **homes and office buildings**.

Examples:

- **Smart Thermostats** – Temperature sensors adjust air conditioning or heating based on room conditions.
 - **Intrusion Detection** – Motion sensors detect unauthorized entry and trigger alarms.
 - **Energy Management** – Light sensors automatically adjust brightness or turn off lights when no one is present.
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Industrial Automation and Safety

Factories use WSNs to improve efficiency, safety, and predictive maintenance.

Examples:

- **Machine Condition Monitoring** – Vibration sensors detect mechanical failures before they happen, reducing downtime.
 - **Gas Leak Detection** – Sensors detect leaks in oil refineries and chemical plants, preventing disasters.
 - **Workplace Safety** – Wearable sensors track worker conditions in hazardous environments (e.g., mines or construction sites).
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Transportation and Traffic Management

Wireless sensors improve road safety, traffic flow, and vehicle management.

Examples:

- **Smart Traffic Lights** – Sensors detect traffic density and adjust signal timings to reduce congestion.
 - **Vehicle Tracking** – GPS-enabled sensors track fleet movements for logistics companies.
 - **Road Condition Monitoring** – Sensors detect potholes and road wear, notifying city authorities for maintenance.
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Smart Grid and Energy Management

WSNs play a crucial role in modernizing **power distribution and energy conservation**.

Examples:

- **Power Grid Monitoring** – Sensors detect faults in electrical grids and prevent blackouts.
- **Smart Meters** – Measure electricity usage and help users optimize energy consumption.

- **Solar Panel Monitoring** – Sensors track solar energy production and alert users about panel efficiency.
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Underwater and Marine Applications

WSNs are used to monitor ocean conditions, marine life, and underwater infrastructure.

Examples:

- **Tsunami Detection** – Ocean-floor sensors detect early signs of tsunamis and send warnings.
 - **Oil Spill Monitoring** – Sensors detect and track oil spills for faster cleanup.
 - **Marine Wildlife Tracking** – Sensors attached to dolphins or sharks track migration patterns.
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Military and Defense

WSNs continue to be important for military operations, security, and surveillance.

Examples:

- **Border Security** – Motion sensors detect unauthorized crossings at national borders.
 - **Missile Tracking** – Wireless sensors track enemy missile launches in real-time.
 - **Supply Chain Monitoring** – Sensors track military equipment shipments and detect tampering.
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Space Exploration

Wireless sensors are used in **satellites, spacecraft, and planetary rovers** to collect data and ensure system health.

Examples:

- **Spacecraft Monitoring** – Sensors track temperature, radiation, and pressure inside spacecraft.
 - **Mars Rover Sensors** – Collect soil and atmosphere data on Mars for research.
 - **Satellite Health Monitoring** – Sensors detect malfunctions in orbiting satellites.
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Retail and Inventory Management

WSNs help businesses **track products, prevent theft, and manage stock efficiently**.

Examples:

- **Smart Shelves** – Sensors detect when products are low and alert store managers.
- **RFID Inventory Tracking** – Wireless tags track product movement in warehouses.

- **Cold Storage Monitoring** – Temperature sensors ensure perishable goods remain at the correct temperature.

Classifications of WSNs

Wireless Sensor Networks (WSNs) are primarily used to **collect and transmit data** from different sensors. The frequency of data collection and transmission plays a crucial role in how these networks function. Based on their **operation mode, functionality, and target applications**, WSNs can be classified into three main types:

1. Proactive Networks

In **proactive networks**, sensor nodes **continuously** monitor their surroundings and transmit data at **regular intervals**. This means the network collects data **even if no major change occurs** in the environment.

Key Features:

- ✓ Sensors wake up periodically to take measurements.
- ✓ Data is sent at fixed time intervals, regardless of changes in the environment.
- ✓ Best for applications that need **continuous monitoring**.
- ✓ Provides a **snapshot of environmental conditions** at every interval.

Examples:

Weather Monitoring: A weather station with temperature, humidity, and wind sensors continuously records and transmits data. Even if there is no change in temperature, the system still updates readings every 10 minutes.

Smart Agriculture: Soil moisture sensors in a farm regularly measure water levels and update the system every hour, ensuring crops get the right amount of water.

Industrial Monitoring: A factory monitors **air quality** and temperature to detect hazardous conditions. Even if everything is normal, data is transmitted every 5 minutes for safety records.

2. Reactive Networks

In **reactive networks**, sensors remain **inactive** until they detect a **significant change** in the environment. Unlike proactive networks, they **do not collect or send data continuously**. Instead, they **react immediately** when a drastic event occurs.

Key Features:

- ✓ Sensors remain in **sleep mode** to save energy.
- ✓ They wake up **only when an event happens** (e.g., a sudden temperature rise or motion detection).
- ✓ Best for **time-sensitive applications** where quick action is required.
- ✓ Conserves energy since data is only transmitted when necessary.

Examples:

Earthquake Detection: Sensors placed in an earthquake-prone area stay **inactive** but activate instantly when strong tremors are detected, sending alerts to authorities.

Fire Alarms: A fire alarm system in a building stays **silent** but triggers an emergency alert when smoke or high temperature is detected.

Security Systems: Motion sensors in a **smart home** remain inactive but send an alert when someone enters the house unexpectedly at night.

3. Hybrid Networks

Hybrid networks **combine both proactive and reactive methods**. Sensors send data at regular intervals (**like proactive networks**) but also react immediately to **sudden changes** (like reactive networks). This approach ensures **continuous monitoring** while also providing **instant alerts** for critical events.

Key Features:

- ✓ Periodic data transmission like proactive networks.
- ✓ Instant response to environmental changes like reactive networks.
- ✓ Balances **energy efficiency** and **real-time responsiveness**.
- ✓ Best for applications that need **both regular updates and quick alerts**.

Examples:

Smart Traffic Monitoring: Traffic sensors send regular updates about vehicle movement but immediately alert authorities if a **major accident or road blockage** is detected.

Healthcare Monitoring: A heart rate monitor in a **hospital** records patient vitals every 5 minutes but **triggers an emergency alarm** if the heart rate drops suddenly.

Wildlife Tracking: Sensors on endangered animals track their location **regularly**, but if an animal **suddenly stops moving** (potential danger), an alert is sent to researchers.

Energy Considerations in WSNs

Since most WSNs rely on **battery-powered sensors**, energy efficiency is crucial. Each type of network must **balance data accuracy and power consumption**:

- **Proactive networks consume more energy** because they continuously collect and send data.
- **Reactive networks save more energy** by only working when necessary.
- **Hybrid networks optimize energy** by using a mix of both approaches.

To **reduce energy consumption**, WSNs use **smart communication protocols** to prevent **data collisions and unnecessary transmissions**. Additionally, energy usage is evenly distributed among all nodes to prevent some sensors from **dying out too quickly**.

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Architecture of Sensor Networks – Simplified Explanation

Wireless Sensor Networks (WSNs) are different from traditional networks because of **their unique applications and communication methods**. The **architecture** of a WSN depends on both individual sensor nodes and the entire network as a whole.

1. Hardware Components of a Wireless Sensor Node

Each **sensor node** consists of the following components:

a) Embedded Microcontrollers

- Sensor nodes use **simple microcontrollers** (like **Atmel** or **MSP 430** from Texas Instruments).
- These microcontrollers must be **energy-efficient** since sensors are battery-powered.
- They should support **different sleep modes** to save power when the sensor is inactive.
- Switching between **active and sleep modes** should be **fast and consume less energy**.

Example:

- In a **wildlife tracking system**, sensors on animals remain in **sleep mode** most of the time. They **wake up only when the animal moves**, reducing battery usage.
-

b) Wireless Communication (Radio Transceivers)

- Sensor nodes use **radio transceivers** to send and receive data.
- Commonly used transceivers include **RFM TR1001** and **Infineon chipsets**.
- Wireless communication methods include **ASK, FSK, and OOK modulation**.
- Researchers are working on **wake-up radio technology**, where sensors can be activated **only when needed**, saving power.

Example:

- In **smart irrigation systems**, soil moisture sensors remain **off** until the control unit **sends a wake-up signal** to check moisture levels.
- Other communication methods like **optical signals (light-based)** or **ultrasound (for underwater applications)** are also being explored.

Example:

- **Underwater sensors** use **ultrasound waves** instead of radio signals to track **ocean currents and marine life**.
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c) Power Supply (Batteries and Energy Harvesting)

- Sensor nodes rely on **batteries** for power.
- **Battery management** is crucial since replacing them is difficult in remote areas.

- Some sensors can use **energy harvesting** (e.g., **solar power**) to recharge themselves.
- **Self-discharge rates and battery lifespan** affect how long a WSN can operate.

Example:

- **Solar-powered weather sensors** in remote locations **store energy during the day** to function at night.
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d) Operating System & Software Management

- WSNs require an **efficient operating system** that balances **low memory usage and flexibility**.
- The system should allow sensors to **communicate, process data, and manage energy** efficiently.
- Techniques like **publish/subscribe models** or **blackboard systems** are used to manage data exchange.

Example:

- A **smart city traffic system** uses **publish/subscribe** methods, where traffic cameras **send updates only when congestion is detected**.

Architecture of a Sensor Network

A **sensor network** is a collection of sensor nodes that work together to monitor physical or environmental conditions such as temperature, humidity, pressure, motion, or pollutants. The architecture of a sensor network is **hierarchical and layered**, ensuring efficient communication and data processing.

1. Layers of Sensor Network Architecture

A typical **Wireless Sensor Network (WSN)** follows a **five-layer** architecture, which helps in organizing its functionalities.

a) Application Layer

- Provides the interface for end-users.
- Handles data visualization, monitoring, and control.
- Examples: Smart homes, industrial monitoring, healthcare systems.

b) Transport Layer

- Ensures reliable data transmission between nodes and base stations.
- Uses **TCP, UDP, or custom lightweight transport protocols**.

c) Network Layer

- Responsible for **routing and forwarding** data packets.
- Uses **multi-hop communication** (data relayed through intermediate nodes).

- Popular routing protocols: **LEACH, AODV, DSR.**

d) Data Link Layer

- Handles **MAC (Medium Access Control) protocols** to prevent collisions.
- Ensures **error detection, data framing, and reliable transmission.**

e) Physical Layer

- Deals with **hardware components** (radio transceiver, antennas).
 - Manages **modulation, signal transmission, and energy efficiency.**
 - Uses **wireless communication standards** like Zigbee, LoRa, Bluetooth, Wi-Fi.
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2. Components of a Sensor Network

A sensor network consists of multiple components working together for efficient sensing and communication.

a) Sensor Nodes

- Small embedded devices with **sensing, processing, and communication** units.
- Examples: Temperature sensors, motion detectors, GPS trackers.

b) Sink Node / Base Station

- Collects data from multiple sensor nodes.
- Acts as a **gateway** between the sensor network and external networks (cloud, servers).

c) Communication Infrastructure

- Defines **how nodes communicate** within the network.
- Uses **single-hop or multi-hop communication.**
- Protocols: Zigbee, LoRa, Wi-Fi, Bluetooth, 6LoWPAN.

d) Power Management System

- Ensures energy-efficient operation.
- Uses **battery-powered nodes or energy-harvesting methods.**

e) Data Processing & Storage System

- Collects and processes sensor data.
- Stores information for further analysis.

A WSN must cover a large area **both for sensing and communication.** The **network density** (number of sensors in an area) is crucial for effective data collection.

a) Lifetime of a WSN

- There is no single rule to define how long a WSN should last.

- Some systems define lifetime as **the failure of one sensor**, while others consider **a certain percentage of failures** acceptable.
- As long as **at least 95% of the monitored area is covered**, the WSN is still useful.

Example:

- In **forest fire detection**, even if **10% of the sensors fail**, the system can still provide accurate fire alerts as long as the remaining **90% are operational**.
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b) Sensor Density and Data Transmission

- Having **more sensors** provides **better coverage**, but too many sensors **increase energy usage and data collisions**.
- An **optimal distance** between sensors should be maintained to **maximize battery life**.
- If sensors are too far apart, **more energy is needed** to send data, reducing lifespan.

Example:

- In a **smart farming system**, sensors placed **too close together** might **send duplicate data**, wasting energy.
 - If they are **too far apart**, communication consumes **more power**, reducing battery life.
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c) Communication Strategies in WSNs

- Data can be transmitted in **two ways**:
 1. **Direct Transmission** – A sensor directly sends data to the **base station**.
 2. **Multi-hop Transmission** – Data is relayed through **intermediate sensors** to reach the base station.

Example:

- In **disaster response systems**, sensors **far from the control center** use multi-hop transmission to **send alerts faster**.
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3. Key Considerations for WSN Network Design

a) Protocol Design for Energy Efficiency

- WSNs need **energy-efficient protocols** that consider both **application requirements and power consumption**.

Example:

- In **smart electricity meters**, sensors **transmit data only when usage changes**, preventing unnecessary power usage.
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b) Quality of Service (QoS) and Reliability

- QoS ensures that data is **delivered accurately and on time**.
- WSNs must handle **data redundancy** (removing duplicate data) to **save energy**.

Example:

- In **patient monitoring systems**, a **heartbeat sensor** should **send an alert immediately** if an irregular pattern is detected, but routine readings can be sent at longer intervals.
-

c) Addressing & Data-Centric Communication

- Traditional networks use **unique addresses (like IP addresses)**, but WSNs often use **data-centric addressing** instead.
- Sensors send data based on **queries like “temperature > 40°C”** instead of sending it from a specific sensor ID.

Example:

- A **smart pollution monitoring system** can be programmed to only send alerts **when air quality falls below safe levels**, reducing unnecessary communication.
-

d) In-Network Processing & Data Aggregation

- Instead of sending raw data, **in-network processing** allows sensors to **process and combine data before transmission**.
- **Data aggregation** reduces **the number of messages sent**, saving energy.

Example:

- In **smart agriculture**, multiple **soil moisture sensors** can send a **combined report** instead of **individual readings** to optimize irrigation.
-

e) Integration with Other Networks (Gateways & Middleware)

- WSNs should be able to **connect to larger networks** (e.g., the Internet).
- A **gateway** is needed to **convert sensor data into a usable format** for other systems.

Example:

- In a **smart home**, sensors send data to a **gateway**, which then updates a **mobile app** to control lights or temperature remotely.
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4. Future Improvements in WSN Architecture

- **Mobile Sensor Nodes** – Some applications may require **moving sensors** to adjust coverage dynamically.

- **Task Reassignment** – WSNs should be able to **change their functions** based on new needs.

Example:

- In **disaster relief operations**, drone-based sensors can be **reprogrammed to track survivors instead of measuring temperature**.

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Physical Layer in Wireless Sensor Networks (WSNs)

What is the Physical Layer?

The **Physical Layer** is the **lowest layer** in the network stack of Wireless Sensor Networks (WSNs). It is responsible for the **actual transmission and reception of data** over a communication medium, such as radio waves, optical signals, or ultrasound.

Functionality of the Physical Layer in WSNs

The Physical Layer in WSNs handles the following key tasks:

1. **Data Transmission & Reception** – Converts digital data into signals for wireless transmission.
2. **Modulation & Demodulation** – Encodes data into radio waves and decodes received signals back into digital data.
3. **Error Detection & Correction** – Ensures reliable communication by handling **signal interference, noise, and packet loss**.
4. **Power Control** – Adjusts transmission power to **reduce energy consumption** and **maximize battery life**.
5. **Multiple Access Techniques** – Manages how multiple sensors share the same communication channel without interference.

Example:

- In **smart agriculture**, sensors monitor **soil moisture** and transmit data to the farmer. The Physical Layer ensures that these sensors can **efficiently send and receive data** over a wireless network without excessive power usage.

Challenges in the Physical Layer for WSNs

1. Energy-Efficient Transmission

Since sensor nodes are **battery-powered**, the main challenge is **how to transmit data while consuming the least energy possible**. Factors affecting energy efficiency include:

- **Overhead Reduction** – Minimizing extra information (such as packet headers) to save energy.
- **Retransmission Reduction** – Avoiding unnecessary retransmissions caused by **signal interference or weak connectivity**.

Example:

- A **wildlife tracking system** places sensors on animals in remote areas. If a sensor **sends redundant data** or has to **retransmit frequently due to interference**, its battery **will drain faster**, reducing the tracking duration.
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2. Modulation Techniques for WSNs

Modulation is the process of **encoding digital data into radio signals** for transmission. Different **modulation techniques** impact energy consumption and communication efficiency in WSNs. Some commonly used techniques include:

a) Code Division Multiple Access (CDMA)

- Allows **multiple sensors to transmit data simultaneously** without interference.
- Useful in **high-density networks** where many sensors operate in the same area.

b) Amplitude Shift Keying (ASK)

- Data is represented by changes in signal **amplitude**.
- Consumes **low power** but is **prone to noise interference**.

c) Frequency Shift Keying (FSK)

- Data is transmitted by **changing the frequency** of the signal.
- More **reliable** than ASK but **uses slightly more power**.

d) On-Off Keying (OOK)

- A simpler version of ASK, where the signal is either **on** or **off** to represent binary data.
- Consumes **low energy**, making it suitable for WSNs.

Example:

- In a **smart parking system**, sensors detect available parking spaces and transmit data to a central server. Using **CDMA**, multiple sensors can communicate at the same time **without interference**, ensuring accurate and real-time updates.
-

3. Communication Standards for WSNs

To improve **energy efficiency and reliability**, industry standards have been developed for WSNs.

IEEE 802.15.4 Standard

- A **widely used** standard for low-power wireless communication in WSNs.
- Forms the basis of **Zigbee**, a popular protocol for IoT and smart home devices.
- Designed to **maximize battery life** while ensuring reliable data transmission.

Example:

- **Zigbee-enabled smart thermostats** use IEEE 802.15.4 to communicate with temperature sensors and adjust room temperature efficiently.
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4. Alternative Transmission Media

Although **radio communication** is the most common in WSNs, some applications require **different communication methods** based on the environment:

a) Optical Communication (Infrared or Laser-Based)

- Used in short-range WSN applications **where obstacles are minimal**.
- Consumes **less power** than radio signals but requires **line-of-sight communication**.

Example:

- In **indoor smart lighting systems**, sensors communicate using **infrared signals** to adjust brightness automatically based on room occupancy.

b) Ultrasound Communication (Underwater WSNs)

- Used in **underwater sensor networks** where radio signals **do not work well**.
- Slower than radio waves but **more reliable** in water environments.

Example:

- In **ocean monitoring**, underwater sensors use **ultrasound waves** to measure temperature and detect pollution levels.
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Future Research Needs in Physical Layer for WSNs

Although some progress has been made in **energy-efficient communication**, more research is needed in:

- **Ultra-low power transmission** methods for increasing battery life.
- **More robust error correction** to minimize retransmissions.
- **Wake-up radio technology**, which allows sensors to **stay in sleep mode until activated**, reducing power consumption.

Example:

- In **smart irrigation**, wake-up radios could allow **soil moisture sensors to remain off until needed**, reducing unnecessary power usage.
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MAC Layer in Wireless Sensor Networks (WSNs)

What is the MAC Layer?

The **Medium Access Control (MAC) Layer** is responsible for **coordinating communication** between sensor nodes in a Wireless Sensor Network (WSN). Since multiple nodes share the same wireless channel, the MAC Layer ensures **efficient data transmission**, **reduces collisions**, and **saves energy**.

Functions of the MAC Layer in WSNs

The MAC Layer in WSNs performs the following key tasks:

1. **Channel Access Control** – Decides when and how a sensor node should transmit data.
2. **Collision Avoidance** – Prevents multiple nodes from transmitting at the same time, which can corrupt data.
3. **Energy Conservation** – Ensures nodes **spend minimal time transmitting and receiving data** to extend battery life.
4. **Data Prioritization** – Determines the order of data transmission, ensuring urgent data is sent first.
5. **Interference Management** – Reduces signal interference by efficiently scheduling transmissions.

Example:

- In a **smart home security system**, motion sensors communicate with a central hub. The MAC Layer ensures that **only one sensor transmits at a time**, avoiding **signal interference** and **reducing energy consumption**.
-

MAC Layer Protocols in WSNs

Active Research in MAC Layer for WSNs

The **MAC Layer** and **Routing Layer** are two of the most actively researched areas in WSNs. A major goal is to **maximize the sleep time of sensor nodes** to save energy. Many MAC protocols use **Time Division Multiple Access (TDMA)**, which assigns specific time slots for transmission, preventing collisions and unnecessary retransmissions.

Wireless Channel as a Broadcast Medium – Explanation

In **Wireless Sensor Networks (WSNs)**, communication between sensor nodes happens **wirelessly** through radio signals. Unlike wired networks, where data is sent through dedicated cables, **wireless communication is broadcast-based**, meaning that **every node within the range of the sender can hear the transmitted data**.

However, not all nodes need the data being sent. If every node processes all messages, it **wastes energy and causes unnecessary interference**. To **optimize communication**, WSNs use different methods to ensure that only the intended recipient processes the message while others **ignore it**.

Types of Channel Allocation in WSNs

1. Static Channel Allocation

- Each sensor node gets a **fixed portion of the bandwidth** (frequency, time, or code).
- Common techniques include:

- **Frequency Division Multiple Access (FDMA)** – Each node gets a specific **frequency range**.
- **Time Division Multiple Access (TDMA)** – Each node gets a **specific time slot** for transmission.
- **Code Division Multiple Access (CDMA)** – Each node uses a unique **code** to send data.

Example:

- In a **weather monitoring network**, sensors measure temperature and humidity at **regular intervals**. Using **TDMA**, each sensor transmits data **in its assigned time slot**, preventing **collisions** and **saving energy**.
-

2. Dynamic Channel Allocation

- The bandwidth is **not pre-allocated**; instead, nodes **compete for the channel** when they need to send data.
- Suitable for networks where the **number of active sensors changes frequently**.
- Based on **Carrier Sense Multiple Access (CSMA)** – Nodes listen to the channel before sending data.

Example:

- In a **disaster response network**, sensors detect gas leaks and **send emergency alerts immediately**. Since all sensors try to send data at once, **collisions may occur**, leading to **data retransmissions and energy loss**.
-

MAC Layer in Hierarchical Clustering Models

- In **hierarchical clustering**, sensor nodes are grouped into **clusters**, each managed by a **Cluster Head (CH)**.
- The **Cluster Head** collects data from its assigned sensors and forwards it to the Base Station (BS).
- To **save energy**, the number of sensors per cluster is kept **small**, and each node transmits data only **in its time slot**.

Example:

- In a **smart city air pollution monitoring system**, sensors in different city zones send data to a **Cluster Head**, which then forwards the data to the control center, reducing **network congestion and energy consumption**.
-

Time Division Multiple Access (TDMA) in WSNs

- **TDMA is useful for both proactive and reactive networks.**

- **Proactive networks:** Sensors transmit data **periodically**, so each node gets a fixed time slot, avoiding **collisions**.
- **Reactive networks:** When an event occurs (e.g., sudden temperature rise), all nearby sensors react at once. Without **TDMA**, they might send data simultaneously, causing **collisions**.

Example:

- In **forest fire detection**, a sudden increase in temperature **triggers multiple sensors**. If they all transmit data at the same time, it may cause **network congestion**. TDMA ensures each node transmits **in an assigned slot**, preventing data loss.
-

Design Issues in MAC Protocols for WSNs

1. Handling Node Failures

- If several sensors **fail** due to battery depletion, the MAC Layer should **find new routes** and **adjust transmission power** to maintain communication.

Example:

- In an **ocean monitoring system**, some sensors might stop working due to harsh conditions. The MAC Layer should **reroute data through alternative paths** to ensure continuous monitoring.
-

2. Energy Consumption in the MAC Layer

Sources of Energy Wastage in WSNs

1. **Collisions** – If two nodes transmit at the same time, data is lost, requiring **retransmission**.
2. **Overhearing** – Nodes receive messages **not meant for them**, wasting energy.
3. **Control Packet Overhead** – Sending control messages consumes energy, especially in **large networks**.
4. **Idle Listening** – Nodes **stay awake even when no data is transmitted**, draining battery power.

Example:

- In a **wildlife tracking system**, sensors should **sleep** when no activity is detected and **wake up only when movement is sensed**, preventing unnecessary power usage.
-

Energy Conservation Techniques in MAC Layer

1. **Sleep Scheduling** – Nodes enter **low-power sleep mode** and wake up only when necessary.
2. **Collision Avoidance** – TDMA ensures each node transmits **without interference**.
3. **Low-Power Listening (LPL)** – Nodes periodically check for incoming messages instead of **staying active continuously**.

Example:

- In a **smart irrigation system**, soil moisture sensors **wake up every 30 minutes** to check water levels and **remain in sleep mode** when not in use.

Link Layer in Wireless Sensor Networks (WSNs)

What is the Link Layer?

The **Link Layer** (also known as the **Data Link Layer**) is responsible for **error detection, packet framing, and reliable data transfer** between sensor nodes in a **Wireless Sensor Network (WSN)**. It ensures that data **sent from one node reaches another node without errors** while **minimizing energy consumption**.

Functions of the Link Layer in WSNs

1. **Packet Framing** – Divides data into **packets** that can be efficiently transmitted.
2. **Error Detection & Correction** – Ensures that **corrupt or lost data** is identified and, if possible, corrected.
3. **Flow Control** – Prevents data loss by ensuring that **nodes do not send more data than the receiver can handle**.
4. **Energy Efficiency** – Optimizes **packet size** and **retransmission mechanisms** to **conserve battery power**.

Example:

- In a **wildlife tracking system**, if a sensor on an animal transmits its location, the **Link Layer** ensures that the data reaches the base station **without corruption**.

Challenges in the Link Layer for WSNs

1. Choosing the Right Packet Size

- **Large packets**: Reduce overhead but **consume more energy** and increase **error probability**.
- **Small packets**: Reduce energy waste from retransmissions but **increase transmission overhead**.
- A balance must be found to **maximize energy efficiency** while ensuring **reliable communication**.

Example:

- In a **smart meter system**, **small packets** are better since the data being sent (electricity usage updates) is **short but frequent**.

2. Energy-Efficient Transmission

- Unlike traditional networks, WSNs **rely on battery-powered sensors**, making **energy efficiency critical**.
- The **Link Layer** must ensure **minimum retransmissions** and **optimized packet sizes** to **extend sensor lifetime**.

Techniques to Reduce Energy Consumption:

- ✓ **Adaptive Packet Size Selection** – Adjust packet size based on **network conditions**.
- ✓ **Low-Power Error Correction** – Use **lightweight error-detection techniques** to **reduce retransmissions**.
- ✓ **Sleep Scheduling** – Allow nodes to **turn off radios** when not in use.

Example:

- In an **ocean monitoring system**, the Link Layer ensures that **wave height sensors** only send data when necessary, reducing power consumption.
-

3. Error Handling & Retransmission Control

- **Wireless communication is prone to errors** due to **interference, noise, and weak signals**.
- The **Link Layer** must **detect errors** and decide whether to **request retransmission** or try to **fix errors automatically**.
- Using **Forward Error Correction (FEC)** allows minor errors to be **corrected without retransmission**, **saving energy**.

Example:

- In **industrial monitoring**, where sensors detect **machine vibrations**, the Link Layer ensures **data integrity** despite interference from nearby machinery.
-

4. Link Layer Reliability & Acknowledgments

- **Acknowledgment (ACK) Mechanisms** ensure that **data is received correctly**.
- **Implicit ACKs** can be used to reduce **extra messages**, saving **energy**.
- **Negative Acknowledgments (NACKs)** notify the sender **only when a packet is lost**, reducing unnecessary transmissions.

Example:

- In a **smart city parking system**, sensors detect **empty spots** and notify the system. **If the message is not acknowledged**, the sensor **retries only if needed**, reducing energy waste.
-

Energy-Efficient Link Layer Protocols for WSNs

1. S-MAC (Sensor-MAC)

How it Works?

- Sensor nodes **follow a fixed schedule** of sleeping and waking up.
- Nodes turn off their radios during the sleep phase to save energy.
- Uses **message passing** to reduce collisions.

Example:

Imagine a **weather station** with multiple sensors collecting temperature data every **10 minutes**. Instead of keeping the radio **on all the time**, S-MAC allows the sensors to sleep and wake up at predefined intervals, **reducing energy consumption**.

Pros: Energy-efficient for periodic data collection.

Cons: May cause **delays** if an event occurs during the sleep period.

2. T-MAC (Timeout-MAC)

How it Works?

- Similar to S-MAC but with a **dynamic sleep-wake schedule**.
- If no data is received within a **timeout**, the node **goes to sleep early**.
- Saves more energy in low-traffic conditions.

Example:

A **forest fire detection system** where sensors stay awake only when they **sense smoke or high temperature**. If no event is detected, nodes sleep early to save power.

Pros: More energy-efficient than S-MAC in variable traffic conditions.

Cons: If the timeout is too short, **important data may be missed**.

3. B-MAC (Berkeley-MAC)

How it Works?

- Uses **Low-Power Listening (LPL)**, where nodes wake up **briefly** to check for signals before sleeping again.
- The sender transmits **long signal** to ensure the receiver is awake to receive data.

Example:

A **smart farming system** where soil moisture sensors wake up occasionally to check if there's any data transmission. If they detect an ongoing signal, they stay awake; otherwise, they go back to sleep.

Imagine **someone knocking on your door**. You wake up for a second, listen for the knock, and go back to sleep if no one is there. The person outside **keeps knocking for a while** to make sure you hear them.

Pros: Provides **low latency** and **high energy efficiency**.

Cons: The **long preamble** transmission increases **overhead** and may cause delays.

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Routing Layer in Wireless Sensor Networks (WSNs) – Simplified Explanation

What is the Routing Layer?

The **Routing Layer** in WSNs is responsible for **finding and maintaining paths** for data to travel from **sensor nodes** to the **Base Station (BS)** or sink. Unlike traditional wireless networks, WSNs have **unique challenges** like **limited battery life**, **frequent topology changes**, and **data redundancy**, which make routing a **critical** component.

Functions of the Routing Layer in WSNs:

1. **Energy-Efficient Routing** – Ensures that data is transmitted using the **least amount of energy** to extend network lifespan.
 2. **Data Aggregation & Processing** – Combines similar data from multiple sensors to **reduce redundant transmissions**.
 3. **Fault Tolerance** – Finds **alternative routes** when sensor nodes fail due to power loss or damage.
 4. **Scalability** – Must handle **thousands of sensor nodes** efficiently.
-

Challenges in Routing for WSNs

1. Flooding-Based Protocols Cause Data Overload

- Traditional routing methods (like **flooding**) send data from every sensor node to the destination **without filtering duplicates**.
- This leads to **high energy consumption** and **wasted bandwidth**.

Example:

- Imagine a **weather monitoring system** with 100 sensors in a city. If **each sensor sends temperature data without checking if others have already sent the same data**, the system gets **flooded with redundant information**, wasting energy and bandwidth.
-

Types of Routing Techniques in WSNs

Routing in WSNs is classified into two main types:

Based on network architecture:

1. **Flat-Based Routing** – Every node has the **same role**, and data is routed based on demand.
2. **Hierarchical-Based Routing** – Nodes are grouped into **clusters**, and one node (Cluster Head) manages communication.
3. **Adaptive Routing** – Routing adjusts based on **network conditions and energy levels**.

Based on protocol operation:

- **Multipath-Based Routing** – Uses **multiple paths** to prevent failures.
 - **Query-Based Routing** – Data is sent **only when requested**.
 - **Negotiation-Based Routing** – Prevents redundant data transmission using **metadata**.
 - **Location-Based Routing** – Nodes use **geographical positioning** to route data efficiently.
-

1.Flat-Based Routing

Directed Diffusion

Directed Diffusion is a popular **flat routing protocol** that is **data-centric** and avoids **unnecessary data transmission** by only sending data that meets a specific **query or interest** from the **Base Station (BS)**.

How Directed Diffusion Works:

1. **Base Station Sends a Query ("Interest")** – The BS **broadcasts a request** for data, such as "Send temperature readings above 35°C".
2. **Sensors Respond & Create Data Paths** – Sensors **that meet the criteria** reply, setting up **gradients (data paths)** toward the BS.
3. **Data Transmission Begins** – The best path is **reinforced**, and duplicate paths are removed to **save energy**.

Example:

- In a **forest fire detection system**, the BS sends an **interest query**: "Which areas have temperatures above 45°C?"
- Only **sensors detecting fire** reply, and the **best data path is selected** to save energy.

2.Hierarchical-Based Routing (Cluster-Based Routing)

What is Hierarchical-Based Routing?

- **Nodes are grouped into clusters**, and each cluster has a **Cluster Head (CH)** that collects data from its **cluster members**.
- **Cluster Heads process and send** data to the Base Station.

Advantages:

- ✓ **Saves energy** – Only **CHs communicate with the Base Station**.
- ✓ **Scalable** – Works well for **large networks**.
- ✓ **Reduces redundant data** – Since **CHs aggregate data** before sending.

Disadvantages:

CHs consume more energy – CHs handle more communication and may **deplete their energy** faster.

Re-clustering overhead – If a CH fails, the network must **reorganize** itself.

Example:

- **Smart City Air Pollution Monitoring**
 - Sensors in different **city zones** measure **pollution levels**.
 - Each **zone** has a **Cluster Head** that collects data from **sensors in that area**.
 - The Cluster Heads then **send summarized data** to the main **control center**.

Protocols Used:

- **LEACH (Low Energy Adaptive Clustering Hierarchy)** – Randomly selects CHs to **balance energy usage**.

Low-Energy Adaptive Clustering Hierarchy (LEACH)

LEACH is a **popular hierarchical routing protocol** that improves **energy efficiency** by rotating **Cluster Heads (CHs)** periodically.

- ✓ CHs are **randomly selected** and changed **periodically** to avoid **battery drain** on a single node.
- ✓ Uses **TDMA & CDMA** to avoid communication collisions.

How LEACH Works:

1. **Setup Phase** – Some sensors **randomly become Cluster Heads (CHs)**.
2. **Steady-State Phase** – CHs collect, compress, and send data to the BS.
3. **Rotation Phase** – After some time, **new CHs are selected** to balance energy usage.

Example:

In a **smart agriculture system**, sensors measure **soil moisture**.

LEACH ensures that the **same sensor doesn't keep acting as CH**, preventing **early battery depletion**.

- **TEEN (Threshold Sensitive Energy Efficient Sensor Network)** – Prioritizes **urgent data** like fire detection.

3. Adaptive Routing

What is Adaptive Routing?

- **Routing dynamically adjusts** based on **network conditions** such as **energy levels** and **node failures**.
- Can switch between **flat-based** and **hierarchical-based** approaches as needed.

Advantages:

- ✓ **Extends network lifetime** – Adapts to energy availability.
- ✓ **Efficient for real-time applications** – Prioritizes **critical data transmission**.

Disadvantages:

More complex – Requires **constant monitoring** of network conditions.

Higher processing requirements – Nodes need to **continuously evaluate network conditions**.

Example:

- **Earthquake Early Warning System**
 - Sensors detect **seismic activity**.
 - If activity is **normal**, data is **sent periodically** (Hierarchical Routing).
 - If a **major earthquake is detected**, all sensors **immediately report** data (Flat-Based Routing).

Protocols Used:

- **SAR (Sequential Assignment Routing)** – Selects routes based on **energy availability and QoS (Quality of Service)**.

Sequential Assignment Routing (SAR)

SAR selects **routes based on three factors**:

1. **Energy Resources** – Avoids low-energy nodes.
2. **Quality of Service (QoS)** – Ensures important data is sent reliably.
3. **Priority Level of Packets** – High-priority data gets faster routing.

✓ **Uses multiple paths** to avoid failures.

✓ **Recomputes routes** if nodes fail.

Example:

- In a **patient monitoring system**, SAR ensures that **critical health data (e.g., heart rate spikes)** is sent **faster and more reliably** than less urgent data (e.g., daily average steps).

4. Multipath-Based Routing

✓ **Uses multiple paths** instead of a **single path** to **increase reliability**.

✓ If one path fails, another **backup path** is used.

✓ **Helps avoid data loss** when nodes fail.

Example:

- **Disaster Response Sensor Network**
 - In a **collapsed building**, sensors detect **survivors**.
 - If the **main communication path is blocked**, the **backup path** ensures data reaches rescuers.

Protocols Used:

- **Braided Multipath Routing** – Uses **alternative paths close to the main route** to save energy.

Braided Multipath Routing

- ✓ Uses "backup" paths that are close to the main route.
- ✓ Less energy-intensive than full multipath routing.

Example:

- In **pipeline monitoring**, multiple sensors monitor **gas leaks**. If one sensor fails, another nearby **backup node** takes over.

5. Query-Based Routing

- ✓ Data is sent only when requested, reducing unnecessary transmissions.
- ✓ The **Base Station (BS)** broadcasts a **query**, and only relevant sensors respond.

Example:

- **Smart Irrigation System**
 - Instead of **all soil moisture sensors** continuously sending data, the **control center** queries them only when needed.

Protocols Used:

- **Directed Diffusion** – BS sends **query requests**, and only relevant nodes **respond**.
-

6. Negotiation-Based Routing

- ✓ Prevents redundant data transmission by using **metadata** to compare new and old data.
- ✓ If a sensor **already sent similar data**, it won't **retransmit**.

Example:

- **Traffic Monitoring System**
 - If a sensor **already sent "Traffic Jam at Location X"**, it will **not resend the same message** unless conditions change.

Protocols Used:

- **SPIN (Sensor Protocols for Information via Negotiation)** – Uses **meta-data exchange** before transmitting data.
-

7. Location-Based Routing

- ✓ Uses **geographical positioning** to route data efficiently.
- ✓ Nodes **closer to the Base Station** forward data instead of using **random paths**.
- ✓ Helps in **saving energy** by reducing **long-distance transmissions**.

Example:

- **GPS-Based Animal Tracking System**
 - Sensors on animals **track their movement** using **GPS**.

- Instead of **sending data to all sensors**, only sensors **closer to the destination** forward the data.

Protocols Used:

- **Geographic Adaptive Fidelity (GAF)** – Puts **unnecessary sensors into sleep mode** to save energy.
 - **Greedy Perimeter Stateless Routing (GPSR)** – Routes data based on **sensor locations**.
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